

has been the largest irrigation system in the world but is also among the more inefficient ones. While about a third of the population does not get safe drinking water, the power of irrigation has turned single-crop lands to four-crop ones, produced paddy in desert areas and grown flowers or spices in semi-arid areas.

Unfortunately, present-day water engineering in India has kept itself preoccupied largely with the promotion of irrigation, hydro-power generation and flood control to some extent. During the 1960s, pumping of groundwater aquifers was promoted for irrigation. The high-priority but low-volume needs of do-

Water redemption:
Safe drinking water is still an issue for victims of last month's Bihar floods



mestic water for drinking and sanitation to 30 per cent of the rural people have still not been met and about 80 per cent of them do not have sanitation. Widespread suffering of the less-privileged from fluorides, arsenic, bacterial contamination or outright industrial pollution do not make news in India. For a country aspiring to be a leading economic power in the world, its inability to provide safe drinking water for all the citizens after 61 years of Independence is not a matter of pride. The success of community initiatives, like the Jal Biradari in Rajasthan, has shown that provisioning of small amounts needed for domestic requirements isn't a problem even in arid areas, when people are involved.

One of the most critical changes needed in India for a more scientific approach to water management is to internalise ecological sciences in the knowledge base for management. Water managers in the government are unable to recognise the fundamental importance of ecological scientific knowledge in their profession. For example, the very first paragraph

in the Report of the Working Group on Water Resources for the 11th Five-Year Plan (2007-2012) of the Ministry of Water Resources takes the position that "... environment and social factors, considerations of sustainability, are the constraints". In the absence of a knowledge base informed by ecology, scientific examination of causes behind extreme events of monsoon inundations can become professionally inadequate, thus opening the path for explanations like "natural disasters", after which there is no accountability. The recent human tragedy in Bihar happened largely due to the failure of the left bank of the Kosi embankment. Open scientific assessment of such failures can produce new knowledge on the complex sediment transfer processes and the drawbacks of engineering in the making of the embankments on such rivers. Branding them "natural disasters" shuts the door for an introspection of the ecological sciences. Politicians find such a branding favourable for distributing money and goodies in the form of relief. The immediate victim of such compromises is the knowledge base for sustainable management of the rivers, of which the long-term and silent victim is the national economy.

The same is the case for water scarcity situations, which are branded "droughts" and for whose relief, funds are approved with great enthusiasm. In India, there are many competent institutions involved in the interdisciplinary study of water management. In addition to the ministerial flights in helicopters over the flooded areas, good governance should be able to put out in public, a comprehensive analysis of the ecology of all the extreme events. For this, open access to hydrological data on the Himalayan rivers would help the nation with much-needed ecological knowledge.

If basic water needs for all are to be met, partnership with communities, independent experts and other private actors should be taken seriously. Water is to be protected from pollution and its recycling should be made mandatory. The efficiency of our vast irrigation system needs to be increased drastically. Further, the institutions for water management should be opened to all stakeholders and doors should be opened for holistic knowledge in water management. Time is running out and the political leadership should remember that water can be as explosive as petroleum, if due attention is not given to it in good time.

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Sinking Feeling

WATER MANAGERS IN THE GOVERNMENT ARE UNABLE TO RECOGNISE THE BASIC IMPORTANCE OF ECOLOGICAL SCIENTIFIC KNOWLEDGE IN THEIR PROFESSION.